Voevodsky Motives

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1 Chow groups

Let k be a perfect field.

Definition 1.1. Given $X \in \mathbf{Sch}_k$, the group of *algebraic r-cycles* on X is the free abelian group on r-dimensional closed integral subschemes Z of X,

$$Z_r(X) = \bigoplus_{Z \subset X} \mathbb{Z} \cdot [Z].$$

The group of *algebraic cycles* is the graded group

$$Z(X) = \bigoplus_{r \geq 0} Z_r(X).$$

An algebraic cycle $\sum_i n_i[Z_i] \in Z(X)$ is *effective* if $n_i \geq 0$ for all i.

Definition 1.2. Given $X \in \mathbf{Sch}_k$ and $W \subset X$ a closed subscheme of pure dimension n over k, with irreducible components $W_1, \ldots, W_s$, the cycle *associated* to W is

$$|W| = \sum_{i=1}^{s} \ell_{\mathcal{O}_{X,W_i}(\mathcal{O}_{W,W_i})} \cdot W_i,$$

where $\ell_{\mathcal{O}_{X,W_i}(\mathcal{O}_{W,W_i})}$ denotes the length of $\mathcal{O}_{W,W_i}$ as an $\mathcal{O}_{X,W_i}$-module.

Definition 1.3. Given $X \in \mathbf{Sch}_k$ and $Z = \sum_i n_i \cdot [Z_i] \in Z(X)$, with all $n_i \neq 0$, the *support* of Z is

$$\mathrm{Supp}(Z) = \bigcup_i Z_i.$$

Definition 1.4. Let D be a Cartier divisor on a scheme $X \in \mathbf{Sch}_k$. Let $Z \subset X$ be an integral subscheme such that $Z \not\subset D$. Then the *intersection product* of D and Z is

$$D \cdot Z = |D \cap Z| = |D \times_X Z|.$$

Writing $Z_r(X)_D \subset Z_r(X)$ for the subgroup generated by r-dimensional integral closed subschemes $Z \subset X$ with $Z \not\subset D$, this operation can be extended linearly to

$$D \cdot (-) : Z_r(X)_D \to Z_{r-1}(X).$$

Definition 1.5. Let $X \in \mathbf{Sch}_k$. Two cycles $Z, Z' \in Z_r(X)$ are *rationally equivalent* if there is a cycle $W \in Z_{r+1}(X \times \mathbb{A}^1_k)_{X \times 0 + X \times 1}$ such that

$$Z - Z' = (X \times 0 - X \times 1) \cdot W.$$

Definition 1.6. Let $X \in \mathbf{Sch}_k$. The *r-th Chow group* of X is the quotient

$$\mathrm{CH}_r(X) = Z_r(X) / \sim_{\text{rat. equiv.}}.$$

2 Chow group operations

Definition 2.1 (projective push-forward)**.** Let $f : X \to Y$ be a projective morphism in $\mathbf{Sch}_k$. For $W \subset X$ an integral closed subscheme of dimension n, the image $f(W) \subset Y$ is also integral and closed, and we have an extension of function fields $f^* : k(f(W)) \to k(W)$. Define the *push-forward of W along f* to be

$$f_*(W) = \begin{cases} 0 & \text{if } \dim_k f(W) < n, \\ [k(W) : k(f(W))] \cdot f(W) & \text{if } \dim_k f(W) = n. \end{cases}$$

By linearity, this extends to $f_* : Z_n(X) \to Z_n(Y)$ and descends to $f_* : \mathrm{CH}_n(X) \to \mathrm{CH}_n(Y)$.

Definition 2.2 (pullback)**.** Let $f : X \to Y$ be a morphism in $\mathbf{Sch}_k$ with X smooth over k. For simplicity, assume X and Y are integral. [TODO: define pullback]

Definition 2.3 (products)**.** Given $X, Y \in \mathbf{Sch}_k$, there is an *external product*

$$\boxtimes : Z_n(X) \otimes Z_m(Y) \to Z_{n+m}(X \times_k Y), \quad W \otimes W' \mapsto W \times_k W'.$$

The product $\boxtimes$ is commutative, associative and unital with respect to $\operatorname{Spec} k \in Z_0(\operatorname{Spec} k)$. It descends to an external product

$$\boxtimes : \mathrm{CH}_n(X) \otimes \mathrm{CH}_m(Y) \to \mathrm{CH}_{n+m}(X \times_k Y).$$

(Define cup product)

Proposition 2.4. (Compatibilities)

3 Correspondences and Chow motives

Definition 3.1. The *category of Chow correspondences* $\mathrm{Cor}_{\mathrm{CH}}(k)$ is the category whose objects are smooth projective schemes over k, and morphisms from X to Y are given by a *correspondence* from X to Y, that is, an element of $\mathrm{CH}_{\dim X}(X \times_k Y)$. The composition of $\alpha \in \mathrm{CH}_{\dim X}(X \times_k Y)$ and $\beta \in \mathrm{CH}_{\dim Y}(Y \times_k Z)$ is given by

$$\beta \circ \alpha = \pi_{XZ*}(\pi^*_{XY}(\alpha) \cdot \pi^*_{YZ}(\beta)),$$

where π_{XY} denotes the projection $X \times_k Y \times_k Z \to X \times_k Y$, et cetera. For any X, the identity morphism id_X is given by the diagonal $\Delta_X(X) \to X$.

Example 3.2. For any morphism $f : X \to Y$ between smooth projective schemes over k, the graph $\Gamma_f = \{(x, y) \in X \times_k Y \mid y = f(x)\} \subset X \times_k Y$ defines a morphism $\Gamma_f : X \to Y$ in $\mathrm{Cor}_{\mathrm{CH}}(k)$. One checks that for morphisms $f : X \to Y$ and $g : Y \to Z$, we have

$$\Gamma_g \circ \Gamma_f = \Gamma_{g \circ f}.$$

Definition 3.3. The *category of effective (homological) Chow motives* $\mathrm{Chow}^{\mathrm{eff}}(k)$ over k is the idempotent completion of $\mathrm{Cor}_{\mathrm{CH}}(k)$, that is,

- the objects are pairs (X, p) with $X \in \mathbf{SmProj}_k$ and $p \in \mathrm{CH}_{\dim X}(X, X)$ such that $p \circ p = p$,
- the morphisms $f : (X, p) \to (Y, q)$ are correspondences $f \in \mathrm{CH}_{\dim X}(X \times_k Y)$ such that $f \circ p = f = q \circ f$.

Remark 3.4. There is a functor

$$m_{\mathrm{CH}} : \mathbf{SmProj}_k \to \mathrm{Chow}^{\mathrm{eff}}(k)$$

which sends $X \in \mathbf{SmProj}_k$ to $m_{\mathrm{CH}}(X) = (X, \Gamma_{\mathrm{id}_X})$ and a morphism $f : X \to Y$ to Γ_f.

4 Finite correspondences and geometric motives

Definition 4.1. Let $X, Y \in \mathbf{Sch}_k$. The group of *finite correspondences* $C(X, Y)$ is the subgroup of $Z(X \times_k Y)$ generated by integral closed subschemes $W \subset X \times_k Y$ such that

(i) the projection $\pi_X : W \to X$ is finite,

(ii) the image $\pi_X(W) \subset X$ is an irreducible component of X.

Lemma 4.2. *Let $X, Y \in \mathbf{Sm}_k$ be irreducible, and $Z \in \mathbf{Sch}_k$, and $W \in C(X, Y)$ and $W' \in C(Y, Z)$. Then each irreducible component C of $S := \mathrm{Supp}(W) \times Z \cap X \times \mathrm{Supp}(W')$ is finite over X and $\pi_X(C) = X$. In particular, the composition of W and W' can be defined*

$$W' \circ W = \pi^S_{XZ*}(\pi^*_{XY}(W) \cdot \pi^*_{YZ}(W')).$$

Definition 4.3. The *category of finite correspondences* is the category $\mathrm{Cor}(k)$ whose objects are objects from $\mathbf{Sm}_k$, and morphisms from X to Y are given by $C(X, Y)$.

Remark 4.4. The fiber product $\times_k$ makes $\mathrm{Cor}(k)$ into a tensor category. Then the bounded homotopy category $\mathbf{K}^{\mathrm{b}}(\mathrm{Cor}(k))$ is a tensor triangulated categoroy.

Definition 4.5. The category $\widehat{\mathrm{DM}}^{\mathrm{eff}}_{\mathrm{gm}}(k)$ is the localization of $\mathbf{K}^{\mathrm{b}}(\mathrm{Cor}(k))$, as a tensor triangulated category, by

- ($\mathbb{A}^1$-*invariance*) For $X \in \mathbf{Sm}_k$, invert $\pi_X : X \times \mathbb{A}^1_k \to X$.
- (*Mayer–Vietoris*) For $X \in \mathbf{Sm}_k$ and Zariski open $U, V \subset X$ with $U \cup V = X$, invert

  $$\mathrm{Cone}(U \cap V \xrightarrow{i} U \oplus V) \xrightarrow{(j_U - j_V)} X,$$

  where i is induced by the inclusions $U \cap V \subset U, V$ and j_U and j_V denote the inclusions $U, V \subset X$.

The *category of effective geometric motives* $\mathrm{DM}^{\mathrm{eff}}_{\mathrm{gm}}(k)$ is the idempotent completion of $\widehat{\mathrm{DM}}^{\mathrm{eff}}_{\mathrm{gm}}(k)$.

Consider the projection $\pi : \mathbb{P}^1_k \to \mathrm{Spec}\, k$ and any section $\sigma : \mathrm{Spec}\, k \to \mathbb{P}^1_k$ and note that $\pi \circ \sigma = \mathrm{id}_{\mathrm{Spec}\, k}$ obviously. Writing $\mathbb{Z} = [\mathrm{Spec}\, k]$ we see that the exact triangle

$$\mathrm{Cone}(\pi)[-1] \longrightarrow [\mathbb{P}^1_k] \underset{\pi}{\overset{\sigma}{\leftrightarrows}} \mathbb{Z} \longrightarrow \mathrm{Cone}(\pi)$$

splits, so that $[\mathbb{P}^1_k] \cong \mathbb{Z} \oplus \mathrm{Cone}(\pi)[-1]$. We define $\mathbb{Z}(1) = \mathrm{Cone}(\pi)[-3]$, so that $[\mathbb{P}^1_k] \cong \mathbb{Z} \oplus \mathbb{Z}(1)[2]$.

Remark 4.6. The above pattern occurs for any smooth scheme X over k with a rational point $x : \operatorname{Spec} k \to X$. The exact sequence

$$\operatorname{Cone}(\pi)[-1] \longrightarrow [X] \underset{\pi}{\overset{\sigma}{\leftrightarrows}} \mathbb{Z} \longrightarrow \operatorname{Cone}(\pi)$$

splits, and we call $\widetilde{[X]} = \operatorname{Cone}(\pi)[-1]$ the *reduced motive* of X.

Definition 4.7. The *category of geometric motives* $\mathrm{DM}_{\mathrm{gm}}(k)$ is defined by inverting the functor $(-) \otimes \mathbb{Z}(1)$ on $\mathrm{DM}_{\mathrm{gm}}^{\mathrm{eff}}(k)$. That is, the objects are $X(n)$ with $X \in \mathrm{DM}_{\mathrm{gm}}^{\mathrm{eff}}(k)$ and $n \in \mathbb{Z}$, and the morphisms are given by

$$\operatorname{Hom}_{\mathrm{DM}_{\mathrm{gm}}(k)}(X(n), Y(m)) = \lim_{N \to \infty} \operatorname{Hom}_{\mathrm{DM}_{\mathrm{gm}}^{\mathrm{eff}}(k)}(X \otimes \mathbb{Z}(n+N), Y \otimes \mathbb{Z}(m+N)).$$

Remark 4.8. The functor $i : \mathrm{DM}_{\mathrm{gm}}^{\mathrm{eff}}(k) \to \mathrm{DM}_{\mathrm{gm}}(k)$ defined by sending X to $X(0)$ is fully faithful. Moreover, the natural map $i(X \otimes \mathbb{Z}(n)) \to X(n)$ is an isomorphism for all $n \geq 0$.

Definition 4.9. There is a functor $M_{\mathrm{gm}} : \mathbf{Sm}_k \to \mathrm{DM}_{\mathrm{gm}}^{\mathrm{eff}}(k)$ sending X to X and f to Γ_f.

5 Nisnevich sheaves with transfers

Definition 5.1. The *category of presheaves with transfer* $\mathrm{PST}(k)$ is the category of additive presheaves of abelian groups on $\mathrm{Cor}(k)$, that is, the category of additive functors $F : \mathrm{Cor}(k)^{\mathrm{op}} \to \mathbf{Ab}$.

The *category of Nisnevich sheaves with transfer on* $\mathbf{Sm}_k$, denoted $\mathrm{Sh}^{\mathrm{Nis}}(\mathrm{Cor}(k))$, is the full subcategory of $\mathrm{PST}(k)$ with objects those F such that, for each $X \in \mathbf{Sm}_k$, the restriction of F to X_{Nis} is a sheaf.

Remark 5.2. In the end, a PST F is a presheaf on $\mathbf{Sm}_k$ together with *transfer maps*

$$\mathrm{Tr}(a) : F(Y) \to F(X)$$

for every finite correspondence $a \in C(X, Y)$, satisfying

$$\mathrm{Tr}(\Gamma_f) = f^*, \quad \mathrm{Tr}(a \circ b) = \mathrm{Tr}(b) \circ \mathrm{Tr}(a), \quad \mathrm{Tr}(a \pm b) = \mathrm{Tr}(a) \pm \mathrm{Tr}(b).$$

Example 5.3. For $X \in \mathbf{Sch}_k$, we have the sheaf with transfers $L(X)$ defined by $L(X)(Y) = \mathrm{Cor}(Y, X)$ for $Y \in \mathbf{Sm}_k$.

Definition 5.4. Let $\mathcal{F}$ be a presheaf of abelian groups on $\mathbf{Sm}_k$. Then $\mathcal{F}$ is called *homotopy invariant* if for all $X \in \mathbf{Sm}_k$, the map

$$\pi_X j : \mathcal{F}(X) \to \mathcal{F}(X \times \mathbb{A}^1_k)$$

is an isomorphism. Furthermore, $\mathcal{F}$ is called *strictly homotopy invariant* if for all $q \geq 0$, the presheaf $X \mapsto H^q(X_{\mathrm{Nis}}, \mathcal{F}_{\mathrm{Nis}})$ is homotopy invariant.

Theorem 5.5. *Let $\mathcal{F}$ be a homotopy invariant PST on $\mathbf{Sm}_k$. Then*

(i) the presheaves $X \mapsto H^q(X_{\mathrm{Nis}}, \mathcal{F}_{\mathrm{Nis}})$ are PST's,

(ii) $\mathcal{F}_{\mathrm{Nis}}$ *is strictly homotopy invariant,*

(iii) $\mathcal{F}_{\mathrm{Zar}} = \mathcal{F}_{\mathrm{Nis}}$ *and* $H^q(X_{\mathrm{Zar}}, \mathcal{F}_{\mathrm{Zar}}) = H^q(X_{\mathrm{Nis}}, \mathcal{F}_{\mathrm{Nis}})$.

Definition 5.6. The category $\mathrm{DM}_-^{\mathrm{eff}}(k)$ is the full subcategory of the derived category $\mathbf{D}^-(\mathrm{Sh}^{\mathrm{Nis}}(\mathrm{Corr}(k)))$ consisting of complexes whose cohomology sheaves are homotopy invariant.

6 Motivic cohomology

Definition 6.1. For $X \in \mathbf{Sm}_k$ and $q \geq 0$, the *motivic cohomology* of X is given by

$$H^p(X, \mathbb{Z}(q)) = \mathrm{Hom}_{\mathrm{DM}^{\mathrm{eff}}_{\mathrm{gm}}(k)}(M_{\mathrm{gm}}(X), \mathbb{Z}(q)[p]).$$

Example 6.2. Applying $\mathrm{Hom}_{\mathrm{DM}_{\mathrm{gm}}(k)}(-, \mathbb{Z}(q)[p])$ to the isomorphism $\pi_{X*} : M_{\mathrm{gm}}(X \times \mathbb{A}^1_k) \to M_{\mathrm{gm}}(X)$ gives an isomorphism

$$\pi_X^* : H^p(X, \mathbb{Z}(q)) \to H^p(X \times \mathbb{A}^1_k, \mathbb{Z}(q)).$$

For $U, V \subset X$ open subschemes, applying $\mathrm{Hom}_{\mathrm{DM}_{\mathrm{gm}}(k)}(-, \mathbb{Z}(q)[p])$ to the distinguished triangle

$$M_{\mathrm{gm}}(U \cap V) \to M_{\mathrm{gm}}(U) \oplus M_{\mathrm{gm}}(V) \to M_{\mathrm{gm}}(U \cup V) \to M_{\mathrm{gm}}(U \cap V)[1]$$

yields the exact sequence

$$\cdots \to H^p(U \cup V, \mathbb{Z}(q)) \to H^p(U, \mathbb{Z}(q)) \oplus H^p(V, \mathbb{Z}(q)) \to H^p(U \cap V, \mathbb{Z}(q)) \to H^{p+1}(U \cup V, \mathbb{Z}(q)) \to \cdots$$

7 Basic computations

Recall that $\mathbb{Z} = M_{\mathrm{gm}}(\operatorname{Spec} k)$ and $\mathbb{Z}(1) = \mathrm{Cone}(\mathbb{P}^1_k \xrightarrow{\pi} \operatorname{Spec} k)[-2]$.

Lemma 7.1. *For all* $n \geq 1$, $M_{\mathrm{gm}}(\mathbb{A}^n_k \setminus \{0\}) \cong \mathbb{Z} \oplus \mathbb{Z}(n)[2n-1]$.

Proof. For $n = 1$, apply the Mayer–Vietoris sequence to the usual covering $\mathbb{P}^1_k = \mathbb{A}^1_k \cup \mathbb{A}^1_k$ to find

$$M_{\mathrm{gm}}(\mathbb{A}^1_k \setminus \{0\}) \to M_{\mathrm{gm}}(\mathbb{A}^1_k) \oplus M_{\mathrm{gm}}(\mathbb{A}^1_k) \xrightarrow{(\,j_1\ -j_2\,)} M_{\mathrm{gm}}(\mathbb{P}^1_k) \to M_{\mathrm{gm}}(\mathbb{A}^1_k \setminus \{0\})[1].$$

Recall that $M_{\mathrm{gm}}(\mathbb{A}^1_k) = M_{\mathrm{gm}}(\operatorname{Spec} k) = \mathbb{Z}$ and $M_{\mathrm{gm}}(\mathbb{P}^1) = \mathbb{Z} \oplus \mathbb{Z}(1)[2]$. Note that $j_1, j_2 : M_{\mathrm{gm}}(\mathbb{A}^1_k) \to M_{\mathrm{gm}}(\mathbb{P}^1_k)$ both factor through $M_{\mathrm{gm}}(\operatorname{Spec} k) = \mathbb{Z}$ as $M_{\mathrm{gm}}(\mathbb{A}^1_k \to \operatorname{Spec} k)$ is an isomorphism. Therefore, it follows that $M_{\mathrm{gm}}(\mathbb{A}^1_k \setminus \{0\}) = \mathrm{Cone}\,(\,j_1\ -j_2\,)\,[-1] = \mathbb{Z} \oplus \mathbb{Z}(1)[1]$.

For $n \geq 2$, we use the covering $\mathbb{A}^n_k \setminus \{0\} = (\mathbb{A}^n_k \setminus \mathbb{A}^{n-1}_k) \cup (\mathbb{A}^n_k \setminus \mathbb{A}^1_k)$, the fact that $(\mathbb{A}^n_k \setminus \mathbb{A}^{n-1}_k) \cap (\mathbb{A}^n_k \setminus \mathbb{A}^1_k) = (\mathbb{A}^{n-1}_k \setminus \{0\}) \times (\mathbb{A}^1_k \setminus \{0\})$, and the isomorphisms $M_{\mathrm{gm}}(\mathbb{A}^n_k \setminus \mathbb{A}^{n-1}_k) \cong M_{\mathrm{gm}}(\mathbb{A}^1_k \setminus \{0\})$ and $M_{\mathrm{gm}}(\mathbb{A}^n_k \setminus \mathbb{A}^1_k) \cong M_{\mathrm{gm}}(\mathbb{A}^{n-1}_k \setminus \{0\})$, in order to obtain the Mayer–Vietoris sequence

$$\begin{aligned}(\mathbb{Z} \oplus \mathbb{Z}(1)[1]) \otimes (\mathbb{Z} \oplus \mathbb{Z}(n-1)[2n-3]) &\to (\mathbb{Z} \oplus \mathbb{Z}(1)[1]) \oplus (\mathbb{Z} \oplus \mathbb{Z}(n-1)[2n-3]) \\ &\to M_{\mathrm{gm}}(\mathbb{A}^n_k \setminus \{0\}) \to (\mathbb{Z} \oplus \mathbb{Z}(1)[1]) \otimes (\mathbb{Z} \oplus \mathbb{Z}(n-1)[2n-3])[1].\end{aligned}$$

The first map can be written down explicitly, so that after expanding brackets it follows that $M_{\mathrm{gm}}(\mathbb{A}^n_k \setminus \{0\}) \cong \mathbb{Z} \oplus \mathbb{Z}(n)[2n-1]$. $\square$

Lemma 7.2. *For all $n \geq 1$, $M_{\mathrm{gm}}(\mathbb{P}^n_k) \cong \bigoplus_{j=0}^n \mathbb{Z}(j)[2j]$.*

Proof. Proof by induction on n, where the case $n = 1$ holds by definition of $\mathbb{Z}(1)$. For $n \geq 2$, we cover $\mathbb{P}^n_k = \mathbb{A}^n_k \cup (\mathbb{P}^n_k \setminus \{0\})$, and note that $\mathbb{P}^n_k \setminus \{0\} = \mathbb{A}^1_k \times \mathbb{P}^{n-1}_k$. The Mayer–Vietoris sequence gives

$$\begin{tikzcd} M_{\mathrm{gm}}(\mathbb{A}^n_k \setminus \{0\}) \arrow[r] \arrow[d, equal, "\wr"] & M_{\mathrm{gm}}(\mathbb{A}^n_k) \oplus M_{\mathrm{gm}}(\mathbb{P}^n \setminus \{0\}) \arrow[r] \arrow[d, equal, "\wr"] & M_{\mathrm{gm}}(\mathbb{P}^n) \arrow[r] & M_{\mathrm{gm}}(\mathbb{A}^n_k \setminus \{0\})[1]. \\ \mathbb{Z} \oplus \mathbb{Z}(n)[2n-1] & \mathbb{Z} \oplus \bigoplus_{j=0}^{n-1} \mathbb{Z}(j)[2j] & & \end{tikzcd}$$

Explicitly writing down the first map, it follows that $M_{\mathrm{gm}}(\mathbb{P}^n) = \bigoplus_{j=1}^n \mathbb{Z}(j)[2j]$. □

In fact, using the Mayer–Vietoris sequence, one can show this local result extends globally.

Theorem 7.3 (Projective bundle formula, [4, Theorem 7.5])**.** *Let $E \to X$ be a vector bundle of rank $n+1$ over $X \in \mathbf{Sm}_k$. Then $M_{\mathrm{gm}}(\mathbb{P}(E)) \cong \bigoplus_{j=0}^n M_{\mathrm{gm}}(X)(j)[2j]$.*

8 Gysin triangles

Theorem 8.1 (Gysin triangle)**.** *Let $i : Z \to X$ be a closed immersion in $\mathbf{Sm}_k$ of codimension n, with open complement $j : U \to X$. Then there is a distinguished triangle*

$$M_{\mathrm{gm}}(U) \xrightarrow{j_*} M_{\mathrm{gm}}(X) \to M_{\mathrm{gm}}(Z)(n)[2n] \to M_{\mathrm{gm}}(U)[1].$$

The case where $Z \to X$ is the zero section of a vector bundle is doable. To prove the general case, one does some stuff with blowups and normal bundles. To reduce to this case.

9 Blowup triangles

(Construct the blowup triangle [6])

10 Motives of schemes of finite type

Definition 10.1. For $X \in \mathbf{Sch}_k$, let $L^c(X)$ be the PST with $L^c(X)(U)$ the free abelian group on irreducible $W \subset U \times_k X$ such that $W \to U$ is quasi-finite, and dominant onto some component of U. Let $C^c_*(X) := C_*(L^c(X))$.

This yields functors

$$C_* : \mathbf{Sch}_k \to \mathrm{DM}^{\mathrm{eff}}_-(k)$$
$$C^c_* : \mathbf{Sch}'_k \to \mathrm{DM}^{\mathrm{eff}}_-(k)$$

where $\mathbf{Sch}'_k$ is the category of schemes over k with proper morphisms.

11 Duality

(Describe the duality operator (if k admits resolutions of singularities))

12 Mixed Tate motives

Definition 12.1. Let k be a field. The *triangulated category of Tate motives* $\mathrm{DMT}(k)_{\mathbb{Q}}$, is the full triangulated subcategory of $\mathrm{DM}_{\mathrm{gm}}(k)_{\mathbb{Q}}$ generated by the Tate objects $\mathbb{Q}(n)$ with $n \in \mathbb{Z}$. Similarly, the *triangulated category of effective Tate motives* $\mathrm{DMT}^{\mathrm{eff}}(k)_{\mathbb{Q}}$ is the full triangulated subcategory of $\mathrm{DM}^{\mathrm{eff}}(k)_{\mathbb{Q}}$ generated by the $\mathbb{Q}(n)$ with $n \geq 0$.

Remark 12.2. The duality operator restricts to $(-)^\vee : \mathrm{DMT}(k)_{\mathbb{Q}} \to \mathrm{DMT}(k)_{\mathbb{Q}}^{\mathrm{op}}$, and sends $\mathbb{Q}(n)$ to $\mathbb{Q}(n)^\vee = \mathbb{Q}(-n)$.

There is a natural weight filtration on $\mathrm{DMT}(k)_{\mathbb{Q}}$ given by

$$\cdots \subset W_{\leq n}\mathrm{DMT}(k) \subset W_{\leq n+1}\mathrm{DMT}(k) \subset \cdots \subset \mathrm{DMT}(k)$$

where $W_{\leq n}\mathrm{DMT}(k)$ is the full triangulated subcategory of $\mathrm{DMT}(k)$ generated by the Tate objects $\mathbb{Q}(-m)$ with $m \leq n$. Note that $W_{\leq n}\mathrm{DMT}(k) = \mathrm{DMT}^{\mathrm{eff}}(k) \otimes \mathbb{Q}(-n)$. Dually, let $W^{>n}\mathrm{DMT}(k)$ be the full triangulated subcategory of $\mathrm{DMT}(k)$ generated by the Tate objects $\mathbb{Q}(-m)$ with $m > n$.

Lemma 12.3. $\mathrm{Hom}_{\mathrm{DMT}(k)}(X, Y) = 0$ *for all* $X \in W_{\leq n}\mathrm{DMT}(k)$ *and* $Y \in W^{>n}\mathrm{DMT}(k)$.

Proof. It suffices to show this for $X = \mathbb{Q}(-a)[s]$ and $Y = \mathbb{Q}(-b)[t]$ with $a \leq n$ and $b > n$, in which case

$$\begin{aligned}\mathrm{Hom}_{\mathrm{DMT}(k)}(\mathbb{Q}(-a)[s], \mathbb{Q}(-b)[t]) &= \mathrm{Hom}_{\mathrm{DMT}(k)}(\mathbb{Q}, \mathbb{Q}(a-b)[t-s]) \\ &= \mathrm{CH}^{a-b}(\operatorname{Spec} k, 2(a-b) - t + s) = 0\end{aligned}$$

as $a - b < 0$. □

One can show (figure out how) that the $W_{\leq n}\mathrm{DMT}(k) \subset \mathrm{DMT}(k)$ are right admissible subcategories, that is, there exist right adjoints $r_n : \mathrm{DMT}(k) \to W_{\leq n}\mathrm{DMT}(k)$ to the inclusions $i_n : W_{\leq n}\mathrm{DMT}(k) \to \mathrm{DMT}(k)$. We let $W_{\leq n}$ be the composition $i_n \circ r_n$. Furthermore, we have maps $\iota_n : W_{\leq n}(X) \to X$ natural in $X \in \mathrm{DMT}(k)$ coming from the counit of the adjunction.

We define $W^{>n}(X) = \mathrm{Cone}(\iota_n)$ which can be shown to be contained in $W^{>n}\mathrm{DMT}(k)$. This yields an exact triangle

$$W_{\leq n}(X) \xrightarrow{\iota_n} X \to W^{>n}(X) \to W_{\leq n}(X)[1]$$

Finally, for any $X \in \mathrm{DMT}(k)$ one defines the *associated graded pieces*

$$\mathrm{gr}_n^W(X) = (W^{>n-1} \circ W_{\leq n})(X) \quad \text{and} \quad \overline{\mathrm{gr}}_n^W(X) = \mathrm{gr}_n^W(X)(-n).$$

Note that $W^{>n-1}W_{\leq n}\mathrm{DMT}(k)$ is equivalent to $\mathbf{D}^{\mathrm{b}}(\mathbf{Vect}_{\mathbb{Q}})$ since $\mathrm{Hom}(\mathbb{Q}(n)[i], \mathbb{Q}(n))$ is 0 for $i \neq 0$ and $\mathbb{Q} \cdot \mathrm{id}$ for $i = 0$. Therefore, we have cohomological functors

$$H^p \circ \mathrm{gr}_n^W : \mathrm{DMT}(k) \to \mathbf{Vect}_{\mathbb{Q}}.$$

13 A t-structure on $\mathrm{DMT}(k)$

Conjecture 13.1 (Beilinson–Soulé vanishing conjecture)**.** *Let k be a field. Then $H^p(k, \mathbb{Q}(q)) = 0$ for $p \leq 0$ except when $p = q = 0$.*

There is a relation between this conjecture and the question whether $\mathrm{DMT}(k)$ is the derived category of an abelian category '$\mathrm{MT}(k)$'. Namely, if we were to have $\mathrm{DMT}(k) = \mathbf{D}^{\mathrm{b}}(\mathrm{MT}(k))$, then

$$H^p(k, \mathbb{Q}(q)) = \mathrm{Hom}_{\mathrm{DMT}(k)}(\mathbb{Q}, \mathbb{Q}(q)[p]) = \mathrm{Ext}^p_{\mathrm{MT}(k)}(\mathbb{Q}, \mathbb{Q}(q)) = 0$$

for $p < 0$. If we assume further that $\mathrm{MT}(k)$ is a rigid tensor category, inducing the tensor product and duality on $\mathrm{DMT}(k)$, with functorial exact weight filtrations W_*, inducing the functors W_n on $\mathrm{DMT}(k)$, and that taking the associated graded with respect to W_* induces a faithful exact functor $\mathrm{gr}^W_* : \mathrm{MT}(k) \to \mathbf{Vect}_{\mathbb{Q}}$. Then, any map $f : \mathbb{Q} \to \mathbb{Q}(q)$ with $q \neq 0$ clearly has $\mathrm{gr}^W_* f = 0$, so that $H^0(k, \mathbb{Q}(q)) = \mathrm{Hom}_{\mathrm{MT}(k)}(\mathbb{Q}, \mathbb{Q}(q)) = 0$ for $q \neq 0$ as well.

Conversely, we have the following theorem.

Theorem 13.2 ([5, Theorem 1.4, Proposition 1.6])**.** *Let k be a field, and suppose the Beilinson–Soulé conjecture holds for k. Then there exists t-structure on $\mathrm{DMT}(k)$ with heart $\mathrm{MT}(k)$ such that:*

(i) $\mathrm{MT}(k)$ *contains all Tate objects $\mathbb{Q}(n)$, and these objects generate $\mathrm{MT}(k)$ as an abelian category, closed under extensions in $\mathrm{DMT}(k)$.*

(ii) The tensor product and duality operator on $\mathrm{DMT}(k)$ restrict to $\mathrm{MT}(k)$, making $\mathrm{MT}(k)$ a rigid tensor category.

(iii) The functors W_n map $\mathrm{MT}(k)$ into itself, giving each object X of $\mathrm{MT}(k)$ a functorial finite weight-filtration

$$0 = W_M(X) \subset W_{M+1}(X) \subset \cdots \subset W_N(X) = X.$$

(iv) Taking associated graded for the weight-filtration gives a faithful exact tensor

$$\mathrm{gr}^W_* = \bigoplus_{n \in \mathbb{Z}} \mathrm{gr}^W_n : \mathrm{MT}(k) \to \mathbf{Vect}_{\mathbb{Q}}$$

making $\mathrm{MT}(k)$ a neutral $\mathbb{Q}$-linear Tannakian category.

(v) There are canonical natural maps

$$\phi_p(X, Y) : \mathrm{Ext}^p_{\mathrm{MT}(k)}(X, Y) \to \mathrm{Hom}_{\mathrm{DMT}(k)}(X, Y[p])$$

for X and Y in $\mathrm{MT}(k)$, which is an isomorphism for $p = 0, 1$ and an injection for $p = 2$.

More concretely, the t-structure is given by $(\mathrm{DMT}(k)_{\leq 0}, \mathrm{DMT}(k)_{\geq 0})$, where $\mathrm{DMT}(k)_{\leq 0} \subset \mathrm{DMT}(k)$ is the full subcategory of all X such that $H^p(\mathrm{gr}^W_n(X)) = 0$ for all $p > 0$ and all n, and $\mathrm{DMT}(k)_{\geq 0} \subset \mathrm{DMT}(k)$ is the full subcategory of all X such that $H^p(\mathrm{gr}^W_n(X)) = 0$ for all $p < 0$ and all n.

Moreover, it was shown that, still assuming the Beilinson–Soulé conjecture, $\mathrm{DMT}(k)$ actually arises as the derived category of $\mathrm{MT}(k)$ [7]. Note that the reference takes k an algebraic field extension over $\mathbb{Q}$ as then the Beilinson–Soulé conjecture holds, as shown by Borel [2].

Theorem 13.3 ([7, Theorem 1.6, Corollary 1.8])**.** *Let k be an algebraic field extension over $\mathbb{Q}$. There exists a canonical t-exact functor*

$$F : \mathbf{D}^{\mathrm{b}}(\mathrm{MT}(k)) \to \mathrm{DMT}(k)$$

which induces the identity on $\mathrm{MT}(k)$. Its composition with the cohomology functor $H : \mathrm{DMT}(k) \to \mathrm{MT}(k)$ associated to the t-structure equals the canonical cohomology functor on $\mathbf{D}^{\mathrm{b}}(\mathrm{MT}(k))$. Furthermore, F is an equivalence of triangulated categories.

(About $\mathrm{MT}(k)$ being a Tannakian category (still assuming BS). Give a description about its Tannaka group $\mathrm{Gal}_{\mathrm{mot}}(k)$ over $\mathbb{Q}$.

Proposition 13.4. $\mathrm{Gal}_{\mathrm{mot}}(k)$ *is a split extension*

$$1 \to U_{\mathrm{mot}}(k) \to \mathrm{Gal}_{\mathrm{mot}}(k) \to \mathbb{G}_m \to 1$$

with $U_{\mathrm{mot}}(k)$ a unipotent group scheme.

Proof. See [4, Lemma 13.3]. □

)

14 Motives of reductive groups

See [1].

15 Motives with finite group actions

Let G be a finite group acting on a scheme X of finite type over k. This induces an action of G on both $M(X)$ and $M^c(X)$, which extends linearly to an action of $\mathbb{Q}[G]$. The goal of this section is to show that this action yields a decomposition of $M(X)$ and $M^c(X)$, which behaves well with respect to multiplication.

For generality, we will work in a idempotent complete K-linear tensor category $\mathcal{A}$ instead of $\mathrm{DM}(k)$.

Definition 15.1. Let $\mathcal{A}$ be a K-linear tensor category, write K for the unit object. For any representation ρ of G defined over K, let $[\rho]$ be $K^{\oplus \dim_K \rho}$ with a natural action of G given by ρ.

Theorem 15.2. *Let G be a finite group with splitting field K. Let $\mathcal{A}$ be an idempotent complete K-linear tensor category, with unit object denoted K. Suppose G acts on an object X of $\mathcal{A}$. Then X decomposes as*

$$X \cong \bigoplus_{\rho} X_\rho \otimes [\rho],$$

where ρ ranges over all irreducible representations of G. The isomorphism is G-equivariant when G acts trivially on X_ρ and naturally on $[\rho]$.

Proof. Write $\rho_1, \ldots, \rho_n$ for the irreducible representations of G. The Artin–Wedderburn theorem gives an isomorphism $K[G] \cong \prod_{i=1}^n \mathrm{Mat}_{d_i \times d_i}(K)$, where $d_i = \dim \rho_i$. This decomposition corresponds to a sequence $e_1, \ldots, e_n \in K[G]$ of pairwise orthogonal central idempotents such that $\sum_{i=1}^n e_i = 1$.

For every $1 \leq j \leq n$, the idempotents e_j induce idempotent morphisms $X \xrightarrow{e_j} X$ which, by assumption, split as

$$X \xrightarrow{r_j} Y_j \xrightarrow{i_j} X$$

with $i_j \circ r_j = e_j$ and $r_j \circ i_j = \mathrm{id}_{Y_j}$. By orthogonality of the idempotents, it follows for $j \neq k$ that $i_j \circ r_j \circ i_k \circ r_k = 0$ and hence $r_j \circ i_k = 0$. Therefore, we have an isomorphism

$$\bigoplus_j Y_j \underset{\prod_j r_j}{\overset{\coprod_j i_j}{\underset{\longleftarrow}{\overset{\longrightarrow}{\sim}}}} X$$

Moreover, since the idempotents e_j are central in $K[G]$, the action of G restricts to Y_j for every j.

Now, every idempotent e_i corresponds to a factor $\mathrm{Mat}_{d_i \times d_i}(K)$. Write $E_{i,j}$ for the matrix with a single one at position (i, j). Hence, we can write $e_i = \sum_{j=1}^{d_i} e_{i,j}$ for pairwise orthogonal idempotents $e_{i,j} = E_{j,j}$ corresponding to the diagonal matrices with a single one at position j (with respect to some basis). This gives a decomposition

$$Y_i \cong \oplus_{j=1}^{d_i} Z_{i,j}$$

Moreover, the $Z_{i,j}$ are isomorphic for all j since the $E_{j,k}$ define an isomorphism between both pieces, with inverse $E_{k,j}$. Hence, $Y_i \cong Z_i \otimes \mathbb{Q}^{\oplus d_i}$. Now all remains is to check that the G-action behaves well, that is, leaves Z_i invariant, which indeed it does (according to the isomorphisms $E_{j,k}$). Therefore,

$$Y_i \cong Z_i \otimes [\rho].$$

□

Remark 15.3. Over $\mathbb{Q}$, we can make a decomposition $X = \bigoplus_\rho X_\rho$, where ρ ranges over the $\mathbb{Q}$-irreducible representations of G, but allows for no further splitting.

Example 15.4. We cannot expect such a decomposition to exist if not every irreducible representation is defined over $\mathbb{Q}$. Let G be a group which does not split over $\mathbb{Q}$, e.g. $G = \mathbb{Z}/3\mathbb{Z}$, and let $\mathcal{A} = \mathbb{Q}[G]$-**Mod**. Let $X \in \mathcal{A}$ be an $\mathbb{Q}$-irreducible representation of G which splits over a splitting field. Then we cannot write $X = Y \otimes [\rho]$ for any Y, because there are no non-trivial idempotents in $\mathrm{End}_{\mathcal{A}}(X)$ since it is a division algebra.

Proposition 15.5. *G finite group acting on X a [...] then, $M(X /\!\!/ G) \cong M(X)^G$*

Proposition 15.6. *Let G act on X and Y, and diagonally on $X \times Y$. Then*

$$(X \times Y)_\rho = \text{natural combination of the } X_\rho \text{ and } Y_\rho.$$

Proof. This is just saying that $[\rho] \otimes [\tau] = [\rho \otimes \tau]$, G-equivariantly. □

16 Motives of toric varieties

Lemma 16.1. *Let X be a scheme of finite type over k. Let $\{U_i\}_{i\in I}$ be an open covering of X such that all intersections $U_{i_1} \cap \cdots \cap U_{i_s}$ are in* $\mathrm{DMT}(k)$*. Then also X is in* $\mathrm{DMT}(k)$*.*

Proof. This is just Mayer–Vietoris applied inductively. □

Proposition 16.2. *Let X be a toric variety Δ. Then $M(X)$ is contained in* $\mathrm{DMT}(k)$*.*

Proof.

- Smooth cones are isomorphic to $\mathbb{A}_k^m \times \mathbb{G}_m^n$ for some m, n, so they are in $\mathrm{DMT}(k)$.
- Smooth toric varieties are in $\mathrm{DMT}(k)$ since they are glued from smooth cones, and all intersections are also smooth cones.
- Any non-smooth toric variety can be blown up in a toric way such that the resulting toric variety is smooth. Furthermore, one can choose the center to be toric as well. From the exact triangle
$$\pi^{-1}(Z) \to \tilde{X} \oplus Z \to X \to \pi^{-1}(Z)[1]$$
now follows that X is also in $\mathrm{DMT}(k)$.

□

Proposition 16.3. *Let X be a smooth toric variety, then $M(X) \cong \bigoplus_i \mathbb{Z}(n_i)[m_i]$ for some $n_i, m_i \in \mathbb{Z}$. The n_i and m_i depend only combinatorially on the data of the fan (i.e. which cones are faces of which cones).*

Corollary 16.4. *Let X be a smooth projective toric variety with fan Δ. Writing $f = \sum_{\sigma\in\Delta}(q-1)^{\operatorname{codim}\sigma} \in \mathbb{N}[q]$, the motive of X is given by*

$$M(X) \cong f(\mathbb{Z}(1)[2]).$$

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(Also, see http://javier.fresan.perso.math.cnrs.fr/mzv.pdf)